

IT 630 — Quiz 1

Fall 2026 • Individual work • 100 points

40 multiple-choice questions • Select one answer, A–E, for each question.

The quiz is designed to take 30 minutes. You may use the full 50-minute class period.

Put your name and student ID on the bubble sheet, and mark every answer there. Work marked on this paper is not graded.

Coverage: material through Monday, September 21 — problem definition and the data lifecycle, reproducibility and provenance, data forms and value types, APIs, table operations, and Git. All scenarios are hypothetical. Each question can be answered on its own; none depends on an earlier answer of yours.

Name: _____	Student ID: _____	Date: _____
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1. A team proposes: “Use Python to analyze our sales spreadsheet.” What is still needed to define the problem?

- A. A faster computer and a newer Python version
- B. A completed analysis before identifying the audience
- C. A chart type and color scheme
- D. The question, intended users, decision, objectives, and scope
- E. A larger spreadsheet, regardless of its contents

2. In the lifecycle used in this course, data management is best described as:

- A. A layer under the cycle that stores and organizes data for retrieval
- B. A sixth stage that runs after data reporting and closes the project
- C. The first half of data preparation
- D. A stage reserved for projects that are hosted in the cloud
- E. Another name for problem definition

3. You are analyzing how weather affects foot traffic at a smoothie shop. Which sequence of early lifecycle stages would you apply first?

- A. Data collection → Data analysis → Problem definition
- B. Data collection → Data preparation → Data reporting
- C. Problem definition → Data preparation → Data reporting
- D. Data analysis → Data reporting → Problem definition
- E. Problem definition → Data collection → Data preparation

4. The problem and the variables needed to answer it were defined correctly, but the team gathered different variables. This is best located as a failure in which stage?

- A. Data collection
- B. Data analysis
- C. Data management
- D. Data preparation
- E. Data reporting

5. After reviewing an analysis, the stakeholder realizes that the original question would not support the decision they need to make. What should the team do?

- A. Keep the question fixed because analysis has already started
- B. Revisit the problem definition and revise the affected later work
- C. Change the chart colors and resend the report
- D. Skip reporting and collect unrelated data
- E. Treat the project as complete because each stage ran once

6. A result is reproducible when:

- A. The analysis produces the same answer after the data is updated
- B. The same analysis steps on the same dataset consistently produce the same answer
- C. The code runs from top to bottom without raising an error
- D. A different team gets a qualitatively similar answer working from a different dataset entirely
- E. The same dataset run through two different pipelines gives the same answer

7. Your pipeline is written in Python. A colleague rebuilds it in R using the same dataset and gets the same answer. Which property have you demonstrated?

- A. Robust
- B. Generalisable
- C. Idempotent
- D. Replicable
- E. Reproducible

8. Which set contains the four things needed to rerun an analysis reproducibly?

- A. The dataset, the author's name, the date it was produced, and the file size in bytes
- B. The results, the conclusions, the citations, and the peer review
- C. A notebook, a virtual machine, a database, and a backup
- D. The code, the data or its retrieval instructions, the environment, and the parameters
- E. The code, a README, a license, and a chart

9. You inherit a project folder containing `analysis.ipynb`, `analysis - Copy.ipynb`, `data.csv`, and `data_new.csv`, and nothing else. Can you reproducibly rerun the reported result from this folder as-is?

- A. Yes, because at least one notebook is present
- B. No, because reproducible work cannot be stored in notebooks
- C. No, because nothing records exactly which code, data, environment, and parameters produced the number
- D. Yes, provided you choose the files whose names look like the most recent versions available
- E. Yes, because both code and data files are present

10. Being able to trace a reported number back to the specific inputs, transformations, and versions that produced it is called:

- A. Normalization
- B. Validation
- C. Aggregation
- D. Deployment
- E. Provenance

11. You hand a project to a new teammate. Which organization best helps them find the entry point and rerun it?

- A. A README explaining the workflow, with code, inputs, and outputs in named folders
- B. Separate copies on several laptops with no shared instructions
- C. A single notebook with the credentials and file paths pasted into its first cell
- D. The final chart plus a screenshot of the notebook
- E. One folder of files named FINAL, FINAL2, and FINAL_new

12. A CSV contains ZIP codes 02138 and 00501. One tool loads them as integers and removes the leading zeros. What best explains what happened?

- A. CSV does not declare column types; the reader inferred an integer type
- B. Pandas cannot read ZIP codes from CSV files
- C. CSV compression removed the characters that did not affect the stored numeric value
- D. Values made of digits alone are quantities rather than identifiers
- E. The CSV schema required each reader to use an integer type

13. A study's analysis uses patient records that data-protection rules forbid from being shared. A colleague concludes that the work therefore cannot be reproducible. What is the best response?

- A. Correct — reproducibility requires the data to be stored in the repository
- B. Reproducibility is a claim about publicly available datasets
- C. It can still be reproducible; the repository records exactly which data was used and how to obtain it
- D. The records may be committed to the repository provided that each file and folder name has first been anonymized
- E. Correct, but the work can still be replicable

14. Parquet is best described as:

- A. A binary format that carries a schema, so column types travel with it
- B. A relational database engine
- C. A compression format for CSV files
- D. A plain-text key-value format
- E. A spreadsheet format that any plain text editor can open and read without conversion

15. Server logs written as JSON, where the fields present vary from record to record, are best classified as:

- A. Semi-structured
- B. Unstructured
- C. Not data
- D. Binary
- E. Structured

16. A CSV table and a free-form interview comment are both saved as text. Which statement is correct?

- A. Binary files are the ones that contain structured data
- B. Both are unstructured because they are stored as text
- C. Text describes the encoding; the CSV can be structured while the comment is unstructured
- D. Saving the comment with a .csv extension gives it a table schema and column headers
- E. Both are structured because a computer can read them

17. A table of reaction-time trials has a trial column holding 1, 2, 3 for each participant. Summing that column gives 12. Why is that number meaningless?

- A. Because the sum was computed over the wrong rows of the trial table
- B. Because there are missing values in the column
- C. Because the column is stored as text
- D. Because trial is an ordered identifier, not a quantity to total
- E. Because integers cannot be summed in pandas

18. A correct column holds True and False. Its mean is 0.833. What is that value?

- A. The number of correct trials
- B. The standard deviation of the column
- C. The accuracy — the proportion of trials that were correct
- D. A meaningless result, since boolean columns cannot be averaged at all
- E. The count of unique values in the column

19. Your repository holds the notebook and README, but a 400 MB order export stays out of it. What single addition keeps the reported number traceable?

- A. The chart produced from the export
- B. A note in the README recording that the export is confidential and large
- C. A provenance line naming the source, period, pull date, and row count
- D. A screenshot of the export's first rows
- E. The export compressed below the repository's file-size limit

20. An API returns the following response:

```
HTTP 429 Too Many Requests
Retry-After: 30
{"error": "too many requests"}
```

Which option correctly identifies its three parts?

- A. 429 and Retry-After: 30 are both body fields; the JSON is a header
- B. Retry-After: 30 is the status, 429 is a header, and the JSON is the body
- C. All three lines belong to the response body
- D. The JSON is the status, 429 is the body, and Retry-After: 30 is a parameter
- E. 429 is the status, Retry-After: 30 is a header, and the JSON is the body

21. Your pull returns HTTP 200 and the body parses cleanly. What has the 200 established?

- A. That no rows are missing
- B. That the request was answered, but not that the data is right or complete
- C. That the data is complete
- D. That the data is correct
- E. That you are permitted to publish the data that the endpoint just returned to you

22. You query a search endpoint. The body contains 20 identifiers, and a field reporting a total of 732 matches. You save the 20 records and write your report. What has gone wrong?

- A. You needed to request a different file format
- B. You received one page of results and treated it as the whole answer
- C. The status code was probably an error
- D. Nothing went wrong; the API returned all the records that it had available to send
- E. The response body was malformed

23. Your script receives HTTP 429 with Retry-After: 30 partway through a loop. What is the best response?

- A. Remove the request parameters so the same loop sends smaller requests
- B. Treat the response as an empty result set and continue the analysis as planned
- C. Retry immediately in a tight loop until one request succeeds
- D. Replace the credentials because 429 means authentication failed
- E. Pause for at least the indicated interval, then retry within the rate-limit policy

24. A shared notebook must call an API requiring a secret key. Which approach follows the lecture's credential-handling guidance?

- A. Commit the key and later add its file to .gitignore
- B. Include the key in the public README for reproducibility
- C. Save the key in the notebook's saved output so classmates can reuse it later
- D. Paste the key into a code cell and hide the cell
- E. Load the key from a protected secret store at runtime, never printing it

25. So that someone else can repeat your pull, the record you write down should name:

- A. The chart you eventually produced from it
- B. The name of the website you visited
- C. The analyst's name, the project deadline, and the folder where it was finally saved
- D. The endpoint, the exact parameters sent, the pull time, and the source version
- E. The number of rows you decided to keep

26. A JSON record for one publication has named fields (doi, type) and a list of three authors. You build a table with one row per author. What happens?

- A. The operation fails because the record is nested
- B. The table has one row, and the authors are lost
- C. The table has three columns, one per author
- D. The table has three rows and the doi field is blank in two of them
- E. The table has three rows, and the named fields repeat down each of them

27. You need the magnitude of the largest individual earthquake. Your table has one row per day containing only the daily mean magnitude and event count. Can that table alone answer the question?

- A. Yes — the largest daily mean equals the largest event magnitude
- B. No — daily means and counts do not identify the largest individual event
- C. Yes — the day with the most events contains the largest earthquake
- D. No — tables cannot hold earthquake measurements
- E. Yes — sorting the daily rows recreates the individual event measurements in order

28. A table contains 100 orders. Exactly 30 have status "paid." You filter to paid orders and then sort by price. What remains?

- A. One row containing the total paid revenue
- B. 100 rows, each representing a price
- C. 30 rows, each representing a customer summary
- D. 70 rows, because the matching rows are removed
- E. 30 rows, each still representing one order

29. You need the highest-priced paid order at the top while preserving all other paid orders for inspection. Which sequence fits?

- A. Sort all orders by price in ascending order, then keep each status
- B. Select only the price column, then discard duplicate prices
- C. Keep only the first paid order before sorting by price
- D. Filter to paid orders, then sort price in descending order
- E. Group by status and replace the orders with their mean prices

30. You have one row per published work, each with a type and publication date. You need a table of counts by type and quarter. Which sequence produces it?

- A. Group by type, then add a quarter column, then count
- B. Lengthen the table, then filter to one quarter, then count
- C. Count all works, then split the total across type and quarter
- D. Widen the table by type, then add a quarter column, then sort by date
- E. Add a quarter column, then group by type and quarter, then count

31. You group a table of 1,000 orders by customer and take the mean order value. There are 240 customers. What does the result look like?

- A. One row holding the overall mean
- B. 240 rows, one per order
- C. 1,000 rows, one per customer
- D. 240 rows, one per customer
- E. 1,000 rows, one per order

32. A wide table has one row for each of four stores and separate sales columns for Monday, Tuesday, and Wednesday. All cells contain a value. You lengthen the three day columns into day and sales columns. What is the result?

- A. 4 rows, one per store
- B. 12 rows, one per store-day
- C. 12 rows, one per store with three separate day columns
- D. 3 rows, one per day across all stores
- E. 7 rows, one per store or day

33. A table has a blank because a measurement for one region-quarter was never supplied. Without additional evidence, how should that blank be interpreted?

- A. All the other measurements in the table should be treated as invalid
- B. The measurement is missing; it must not be read as a measured zero
- C. That region cannot exist
- D. The missing value can be taken as equal to the largest value observed elsewhere
- E. The measurement was exactly zero

34. An orders table has 50 rows containing a region code. A lookup table contains exactly one name for every code used by the orders. You join the names onto the orders by region code. What should the result contain?

- A. 50 order rows with the matching region names added
- B. Only the lookup rows, with the orders discarded
- C. A sorted table even though no sorting was requested
- D. One row per region, with the orders automatically totaled
- E. 50 extra rows containing only region names

35. You edit report.py and README.md, then run `git add report.py`. Before making any further edits, what will a plain `git commit` record?

- A. No changes at all until both of the files have been pushed
- B. The entire shared remote repository
- C. Both edits, because commit saves each file in the folder
- D. The staged report.py change, leaving the README.md edit uncommitted
- E. Only README.md because report.py is already staged

36. A report's result changed between two known commits. Which command shows the changed lines in report.py between those commits?

- A. `git add report.py`
- B. `git status`
- C. `git diff OLD NEW -- report.py`
- D. `git log --oneline OLD..NEW report.py`
- E. `git push`

37. You use `git restore --source=GOOD_COMMIT report.py` to recover a known-good file and verify that the report is correct. What is needed to record this repair in history?

- A. Nothing; restore automatically creates a commit
- B. Stage and commit the restored file as a new change
- C. Clone the repository again so that the restored file is saved somewhere
- D. Delete all commits after GOOD_COMMIT
- E. Add report.py to .gitignore

38. You committed a file that contained an API key, noticed, and added the file to .gitignore. Is the key out of the repository?

- A. Yes, once you commit the .gitignore change
- B. Yes — .gitignore removes the file from the project
- C. No — .gitignore never untracks an already tracked file or erases existing history
- D. No, but that depends on whether you had already pushed before ignoring it
- E. Yes, because ignored files are excluded from storage

39. Before letting an agent edit a project, you commit your own changes, confirm a clean working tree, and create a separate branch. After it finishes, what should you do before keeping its work?

- A. Check whether the agent reported that it succeeded, then keep the work
- B. Merge immediately, since working on a branch is what makes the result correct
- C. Force-push its branch over main without inspecting it
- D. Inspect status and diff for the requested change and every unrelated file
- E. Ignore untracked files because they cannot contain generated data

40. You made three commits. `git status` says the working tree is clean and your branch is three commits ahead of origin/main. A teammate clones origin/main and cannot see the commits. Why, and what fixes it?

- A. A clean working tree erased the commits; stage the files again
- B. Commit messages are private unless copied into the README
- C. The commits are still local; push the branch to the shared remote
- D. Cloning omits all but the newest commit; the teammate should run `git diff`
- E. Three commits cannot be shared until they are combined into one commit

End of quiz — check that every answer is marked on the bubble sheet.